Lecture 11 recap

In Lecture 11:

- Learnt the existence of the inverse Laplace transform and its linear properties
- Learnt to calculate LTs and ILTs using the formulae "cheat" sheet + linearity
- Learnt to use partial fractions to calculate ILTs of suitable functions

Lecture 12

In Lecture 12, we will:

1. Learn how to express the LT of derivatives in a convenient form
2. Learn how to use the LT to solve IVPs
3. Learn the s-shifting and t-shifting theorems to calculate LTs and ILTs

Theorem (Laplace transform of the derivative)

Let f be a differentiable function on $[0, \infty)$ of exponential order and f' piecewise continuous. In this case $\mathscr{L}\{f\} = F$ exists and

$$\mathscr{L}\{f'\} = s\,F(s) - f(0)$$

The initial condition is encoded into the Laplace transform!

Proof:

$$\begin{aligned}
\mathscr{L}\{f'\} &= \int_0^\infty \mathrm{e}^{-st} f'(t)\ \mathrm{d}t = \lim_{b\to\infty} \int_0^b \mathrm{e}^{-st} f'(t)\ \mathrm{d}t \\
&= \lim_{b\to\infty} \left\{ \left[\mathrm{e}^{-st} f(t)\right]_0^b - \int_0^b -s\,\mathrm{e}^{-st} f(t)\ \mathrm{d}t \right\} \\
&= -f(0) + s \lim_{b\to\infty} \int_0^b \mathrm{e}^{-st} f(t)\ \mathrm{d}t = s\,F(s) - f(0)
\end{aligned}$$

as required

Application to ODEs; **Exercise 3.11**: Solve $y'(t) - y(t) = 0$, $y(0) = 1$

Take Laplace transforms of both sides of the ODE:

$$\mathscr{L}\{y' - y\} = \mathscr{L}\{0\}$$

$$\Rightarrow \quad \underbrace{sY - y(0) - Y}_{(s-1)\,Y - 1} = 0$$

$$\Rightarrow \quad (s-1)\,Y = 1 \quad \Rightarrow \quad Y(s) = \frac{1}{s-1}$$

Take the inverse Laplace transform

$$\mathscr{L}^{-1}\{Y\} = \mathscr{L}^{-1}\left(\frac{1}{s-1}\right) = \mathrm{e}^t \quad \text{by LT table}$$

Theorem (Higher order derivatives)

Let f be an n-fold differentiable function on $[0, \infty)$ where all the derivatives up to order $n-1$ are of exponential order and $f^{(n)}$ is piecewise continuous. Then

$$\mathscr{L}\left\{f^{(n)}\right\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \ldots - f^{(n-1)}(0)$$

This can be proved using integration by parts and induction.

We will use the case of a second derivative extensively:

$$\mathscr{L}\{f''\} = s^2 F(s) - s f(0) - f'(0)$$

Exercise 3.12: $y''(t) - 3y'(t) + 2y(t) = 0$ where $y(0) = 0$ and $y'(0) = 2$

$$\mathscr{L}\{y'' - 3y' + 2y\} = \mathscr{L}\{0\}$$

$$\Rightarrow \quad \mathscr{L}\{y''\} - 3\,\mathscr{L}\{y'\} + 2\,\mathscr{L}\{y\} = 0$$

$$\Rightarrow \quad s^2\,Y - s\,y(0) - y'(0) - 3\left\{s\,Y - y(0)\right\} + 2\,Y = 0$$

$$\Rightarrow \quad \left(s^2 - 3\,s + 2\right)Y - 2 = 0$$

Therefore

$$Y = \frac{2}{s^2 - 3s + 2} = \frac{2}{(s-1)(s-2)}$$

$$\Rightarrow \quad y(t) = \mathscr{L}^{-1}\{Y(s)\} = 2(\mathrm{e}^{2t} - \mathrm{e}^{t}) \qquad \text{(Ex 3.10)}$$

Exercise 3.13: Find the Laplace transform of $\mathrm{e}^{3\,t}\,t^2$

Theorem (s-shift)

If $\mathscr{L}\{f\}$ exists for $\mathrm{Re}(s) > \alpha$, then

$$\mathscr{L}\{\mathrm{e}^{\omega\,t} f(t)\} = F(s-\omega)\,, \qquad \mathrm{Re}(s-\omega) > \alpha$$

$$\mathscr{L}\{t^2\} = \frac{2}{s^3} \quad \text{for} \quad \mathrm{Re}(s) > 0$$

$$\Rightarrow \quad \mathscr{L}\{\mathrm{e}^{3t}\,t^2\} = \frac{2}{(s-3)^3} \quad \text{for} \quad \mathrm{Re}(s) > 3 \quad \text{using } s\text{-shift}$$

Exercise 3.14: Find the inverse Laplace transform of $\dfrac{3}{(s+2)^2+9}$

$$\mathscr{L}^{-1}\left\{\frac{3}{s^2+9}\right\} = \sin 3t \qquad \text{from LT table}$$

$$\Rightarrow \quad \mathscr{L}^{-1}\left\{\frac{3}{(s+2)^2+9}\right\} = \mathrm{e}^{-2t}\sin 3t \qquad \text{by } s\text{-shifting}$$

Where is the shifted transform valid?

Discontinuous functions

Definition (Heaviside step function)

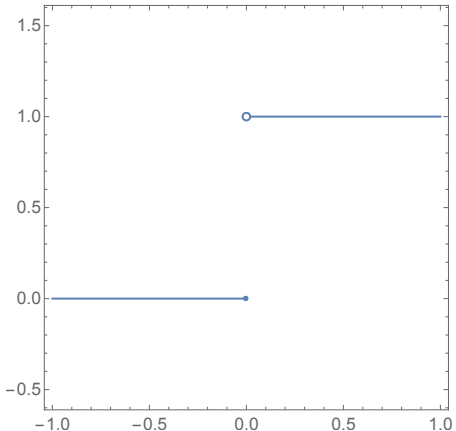
$$H(t) = \begin{cases} 0 & t \leqslant 0 \\ 1 & t > 0 \end{cases}$$

The value of $H(0)$ is usually irrelevant.

It can be translated so the discontinuity occurs at $t = c$. This results in

$$H_c(t) = H(t - c)$$

It can be used to model turning processes on or off.



Exercise 3.15: Find the Laplace transform of $f(t) = \sin(t-3)H(t-3)$

Theorem (t-shift)

If $F(s)$ exists for $\mathrm{Re}(s) > \alpha$, and $c \geqslant 0$, then

$$\mathscr{L}\Big\{ f(t-c)H(t-c) \Big\} = \mathrm{e}^{-cs} F(s) \,, \quad \mathrm{Re}(s) > \alpha$$

$$F(s) = \mathrm{e}^{-3s} \mathscr{L}(\sin t)$$

$$\Rightarrow \quad F(s) = \frac{\mathrm{e}^{-3s}}{s^2+1} \quad \text{for} \quad \mathrm{Re}(s) > 0 \qquad \text{by } t\text{-shifting}$$

The factor e^{-3s} signals a delay of $\sin(t)$ by 3 units of time.

Exercise 3.16: Find the ILT of $\dfrac{\mathrm{e}^{-2s}}{s-4}$ where $\mathrm{Re}(s) > 4$

$$\mathscr{L}^{-1}\left\{\frac{1}{s-4}\right\} = \mathrm{e}^{4t} \quad \text{for} \quad \mathrm{Re}(s) > 4$$

$$\Rightarrow \quad \mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s-4}\right\} = \mathrm{e}^{4(t-2)}H(t-2) \qquad \text{by } t\text{-shifting}$$

The factor e^{-2s} signals a delay of $f(t)$ by 2 units of time.